

Exercise :

1) Given $W = \{\vec{x} \in \mathbb{R}^3 : 2x_1 - x_2 + 3x_3 = 0\}$ find a matrix M s.t. $W^\perp = \text{null } M$

Setting $x_2 = 2x_1 + 3x_3$ we have that

$W = \text{span} \left(\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix} \right)$. Now as

$$\begin{aligned} x_2 &= 2x_1 + 3x_3 \\ x_1 &= x_1 \\ x_3 &= x_3 \end{aligned}$$

$$\vec{x} = x_1 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$$

$$W = \text{range } A \text{ where } A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \\ 0 & 1 \end{bmatrix}$$

we have that by Theorem 5.9

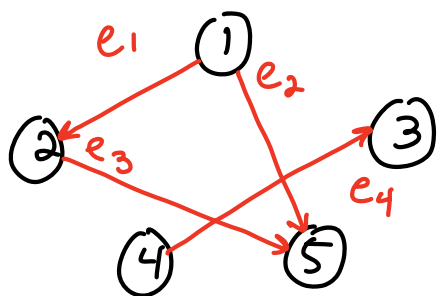
$$W^\perp = \text{null } A^T \text{ or } W^\perp = \text{null } M$$

$$\text{where } M = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \end{bmatrix}$$

Directed Graphs and Incidence Matrices
(See chapter 10 in Strang)

Def 6.1 A directed graph consists of a set of vertices or nodes, and a set of directed edges.

Example: A diagram of a directed graph with 5 nodes and 4 edges is given by



Incidence matrix

Def 6.2 Given directed graph G with n nodes and m edges, the associated incidence matrix denoted $A(G)$ is an $m \times n$ matrix where

$$A(G)_{i,j} = \begin{cases} +1 & \text{if edge } i \text{ goes into node } j \\ -1 & \text{if edge } i \text{ comes out of node } j \\ 0 & \text{if edge } i \text{ avoids node } j \end{cases}$$

Example: Given the graph G from above

$$A(G) = \begin{matrix} & \begin{matrix} ① & ② & ③ & ④ & ⑤ \end{matrix} \\ \begin{matrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{matrix} & \begin{bmatrix} -1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 & 0 \end{bmatrix} \end{matrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} x_2 - x_1 \\ x_5 - x_1 \\ x_5 - x_2 \\ x_3 - x_4 \end{bmatrix}$$

Fundamental Subspaces of $A(G)$ and interpretations

i. range $A(G)$: let $\vec{x}$ represent the voltage applied to each node. Then $A(G) \cdot \vec{x}$ represents the differences across each edge

Kirchoff's Voltage Law (KVL): Differences across a loop sum to 0.

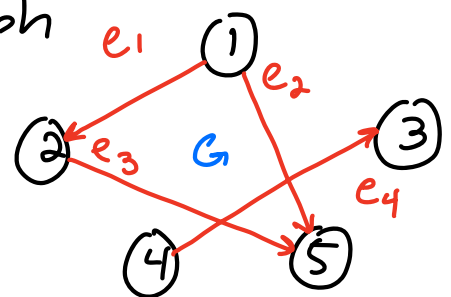
ii. null $A(G)$: $A(G) \cdot \vec{x} = \vec{0}$ when the differences are all 0. The nullity of $A(G)$ is equal to the number of connected components.

iii. range $A(G)^T$: let $\vec{y}$ represent the current applied to each edge. Then $A(G)^T \cdot \vec{y}$ represents the net flow (flow in - flow out) at each node

Kirchoff's Current Law (KCL): The algebraic sum of currents at each node is 0.

iv. null $A(G)^T$: The nullity of $A(G)^T$ is the number of (reduced) loops in G .

Example: Given the directed graph



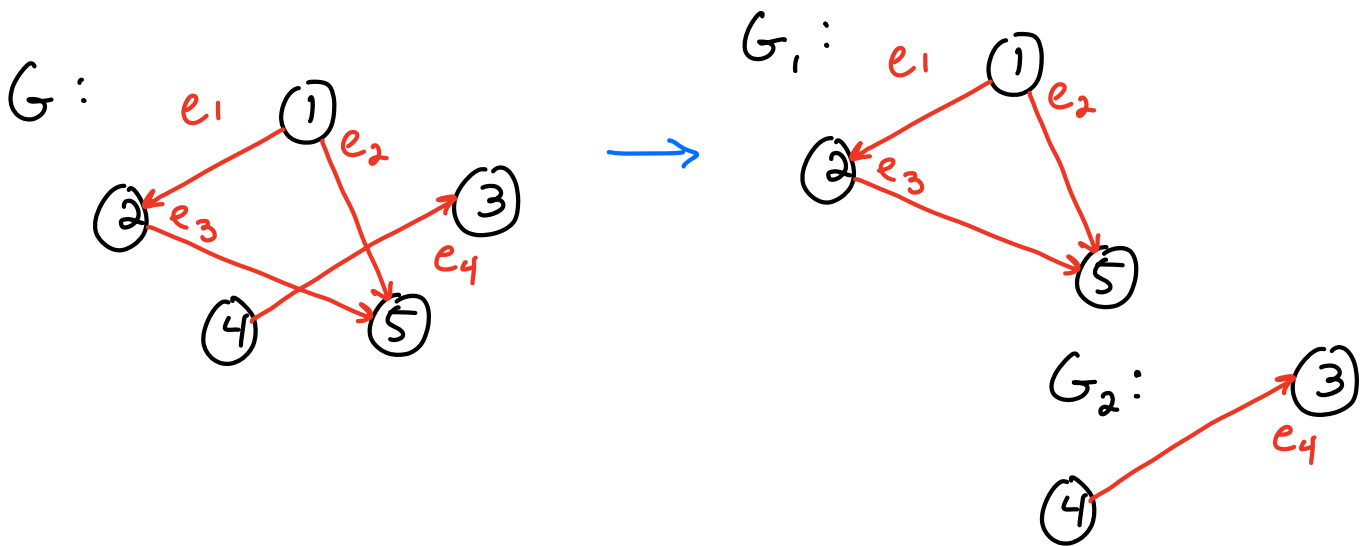
we have a loop by following the edges:

$e_1 + e_3 - e_2$ and thus and thus the

vector $\begin{matrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{matrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} \in \text{null } A(G)^T$

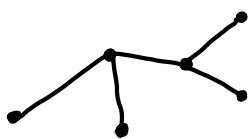
$e_1 - e_2 + e_3 + 0e_4 = \vec{0}$

Def 6.3 A graph is connected if there exists a path, i.e. a set of edges, between each node. If G is disconnected then we write $G = \bigoplus_{i=1}^k G_i$ where each G_i is a connected component of G .



Here $G = G_1 \oplus G_2$ ($A(G) = A(G_1) \oplus A(G_2)$)

Def 6.4 A graph with no loops is a tree



Theorem 6.5 let $G = G_1 \oplus \dots \oplus G_k$ and let $A = A(G)$. Then

i. $\text{range } A = \text{span of columns of } A \text{ where one node is deleted from each } G_i$

ii. $\text{range } A^T = \text{Span of rows of } A \text{ corresp. to the maximal tree of } A$
(delete one edge from each simple loop)

iii. $\text{null } A = \text{span}(\vec{x}_1, \dots, \vec{x}_k)$ where

$$(\vec{x}_i)_j = \begin{cases} 1 & \text{if node } j \text{ is in } G_i \\ 0 & \text{if node } j \text{ is not in } G_i \end{cases}$$

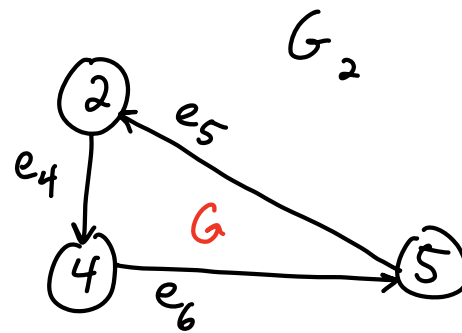
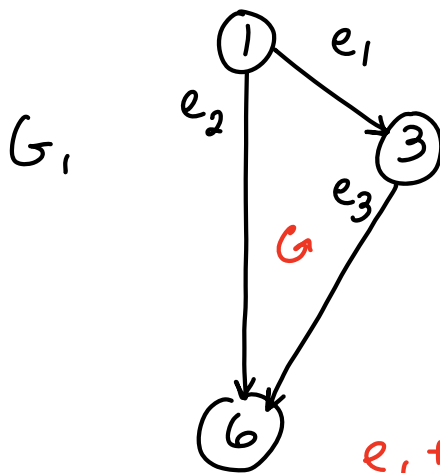
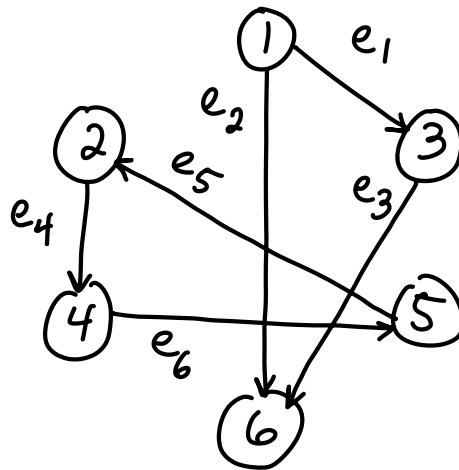
iv. $\text{null } A^T = \text{Span of loop currents (vectors)}$

Exercise:

2) Given the incidence matrix

$$A = \begin{bmatrix} -1 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \end{bmatrix}$$

Sketch the associated directed graph and find bases for the fundamental subspaces.



$$e_1 + e_3 - e_2 = 0$$

$$e_4 + e_6 + e_5 = 0$$

$$e_1 + e_3 = e_2$$

Looking at the connected components we have that

$$\text{null } A = \text{Span} \left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right)$$

and looking at the loops

$$\text{null } A^T = \text{span} \left(\begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right)$$

Now deleting one node from each connected component we have that (5,6)

$$\text{range } A = \text{span} (A_{\cdot,1}, A_{\cdot,2}, A_{\cdot,3}, A_{\cdot,4})$$

and deleting one edge from each loop we have that (e_3, e_6)

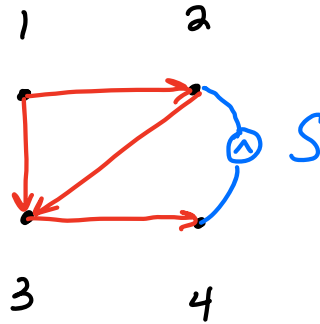
$$\text{range } A^T = \text{span} (A_{1,\cdot}, A_{2,\cdot}, A_{4,\cdot}, A_{5,\cdot})$$

Ohm's Law

Let $\vec{x}$ be the voltages at each node and set $\vec{z} = A\vec{x}$ to be the potential differences. Then letting $\vec{y}$ denote the current across each edge we have that $\vec{y} = C\vec{z}$ where C is given by Ohm's Law. Now absent a source like a battery, KCL gives us

that $A^T C A \vec{x} = \vec{0}$. Adding $\vec{f}$ the balance equation becomes $A^T C A \vec{x} = \vec{f}$.

Example: Assume no resistance so that $C = I$
Now given the circuit



with source S going into node 2. Now we have that

$$A^T A = \begin{bmatrix} -1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

which is not invertible. So we ground node 4 (set $x_4 = 0$)

$$A^T A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 3 \end{bmatrix}$$

and now $A^T A \vec{x} = \vec{f}$ where $\vec{f} = \begin{bmatrix} 0 \\ S \\ 0 \end{bmatrix}$

is solved by

$$\left[\begin{array}{ccc|c} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & s \\ -1 & -1 & 3 & 0 \end{array} \right]$$

$$\vec{x} = \begin{bmatrix} 4s/3 \\ 5s/3 \\ s \end{bmatrix}$$

and the current is given by

$$\begin{aligned} \vec{y} &= -A\vec{x} = - \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 4s/3 \\ 5s/3 \\ s \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} -s/3 \\ s/3 \\ 2s/3 \\ s \end{bmatrix} \end{aligned}$$